HW6 12209095 杜奕诚

1. (a) CML $E(\tilde{r}_c) = r_f + \frac{E(\tilde{r}_m) - r_f}{\sigma_m}\sigma_c$

$E(\tilde{r}_m) = 10\%$ $\sigma_m = 40\%$ $r_f = 2\%$

$E(\tilde{r}_c) = 2\% + \frac{10-2}{40}\sigma_c = 2\% + \frac{1}{5}\sigma_c$

(b) $8\% = 2\% + \frac{1}{5}\sigma_c$ $\sigma_c = 30\%$

$w \cdot 2\% + (1-w)\,10\% = 8\%$

$\therefore w = 0.25$

\$250 on risk-free asset, \$750 on tangent portfolio

2. (a) SML $E(\tilde{r}_c) = r_f + [E(\tilde{r}_m) - r_f]\cdot\beta_c$

$= 3\% + 9\%\cdot\beta_c$

(b) $E(\tilde{r}_i) = 3\% + 9\% \times 0.6 = 8.4\%$

(c) $\beta_j = \frac{Cov(\tilde{r}_j, \tilde{r}_m)}{Var(\tilde{r}_m)} = \frac{Corr(\tilde{r}_j, \tilde{r}_m)\,\sigma_m\sigma_c}{\sigma_m^2} = \frac{25\% \times 60\%}{30\%} = 0.5$

$E(\tilde{r}_j) = 3\% + 9\% \times 0.5 = 7.5\%$

(d) $\beta_j = \frac{-25\% \times 80\%}{30\%} = -\frac{2}{3}$

$E(\tilde{r}_j) = 3\% + 9\% \times -\frac{2}{3} = -3\%$

3. $E(\tilde{r}_j) = r_f + [E(\tilde{r}_m) - r_f]\,\beta_j$

$= 2\% + 6\%\,\beta_j$

Intel: $\beta_j = 1.05$, $E(\tilde{r}_j) = 8.3\%$

Ford Motor: $\beta_j = 1.25$ $E(\tilde{r}_j) = 9.5\%$

Delta Airlines: $\beta_j = 1.30$ $E(\tilde{r}_j) = 9.8\%$

US Steel: $\beta_j = 1.90$ $E(\tilde{r}_j) = 13.4\%$

Procter & Gamble: $\beta_j = 0.65$ $E(\tilde{r}_j) = 5.9\%$

Wal-Mart: $\beta_j = 0.70$ $E(\tilde{r}_j) = 6.2\%$

AT&T: $\beta_j = 0.75$ $E(\tilde{r}_j) = 6.5\%$

Pfizer: $\beta_j = 0.90$ $E(\tilde{r}_j) = 7.4\%$